

Quản trị Hadoop

- Ubuntu Server 22.04

- Hadoop 3.3.4

- Login với vai trò root (pass: root) để thực hiện những công việc sau

1. Khởi chạy Hadoop

```
$ su hadoopkhai
```

```
$ cd ~
```

```
$ start-dfs.sh
```

2. Khởi chạy yarn để dùng map reduce

```
$ start-yarn.sh
```

3. (hdfs) Kiểm tra tình trạng của hệ thống hadoop file system

```
$ hdfs fsck /
```

```
hadoopkhai@khai-master: ~  
/user/hadoopkhai/input1/file2.txt: Under replicated BP-484601954-192.168.111.1-1740328187673:blk_107374:  
eplica(s), 0 decommissioning replica(s).  
  
Status: HEALTHY  
Number of data-nodes: 1  
Number of racks: 1  
Total dirs: 74  
Total symlinks: 0  
  
Replicated Blocks:  
Total size: 601350 B  
Total files: 8  
Total blocks (validated): 8 (avg. block size 75168 B)  
Minimally replicated blocks: 8 (100.0 %)  
Over-replicated blocks: 0 (0.0 %)  
Under-replicated blocks: 8 (100.0 %)  
Mis-replicated blocks: 0 (0.0 %)  
Default replication factor: 2  
Average block replication: 1.0  
Missing blocks: 0  
Corrupt blocks: 0  
Missing replicas: 8 (50.0 %)  
Blocks queued for replication: 0  
  
Erasure Coded Block Groups:  
Total size: 0 B  
Total files: 0  
Total block groups (validated): 0  
Minimally erasure-coded block groups: 0  
Over-erasure-coded block groups: 0  
Under-erasure-coded block groups: 0  
Unsatisfactory placement block groups: 0  
Average block group size: 0.0  
Missing block groups: 0  
Corrupt block groups: 0  
Missing internal blocks: 0  
Blocks queued for replication: 0  
FSCK ended at Sun Mar 02 07:37:30 UTC 2025 in 30 milliseconds  
  
The filesystem under path '/' is HEALTHY  
hadoopkhai@khai-master:~$
```

4. Mở trình duyệt để xem tình trạng hadoop server chạy

<http://localhost:9870/dfshealth.html#tab-overview>

localhost:9870/dfshealth.html#tab-overview

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities

Overview 'khai-master:9000' (✓active)

Started:	Sun Mar 02 07:33:30 +0000 2025
Version:	3.3.4, ra585a73c3e02ac62350c136643a5e7f6095a3dbb
Compiled:	Fri Jul 29 12:32:00 +0000 2022 by stevel from branch-3.3.4
Cluster ID:	CID-b791e412-bdb1-4d7c-8ce7-b283cb43f3b9
Block Pool ID:	BP-484601954-192.168.111.1-1740328187673

5. Xem các tập tin thư mục đang lưu trữ trên hệ thống hdfs

<http://localhost:9870/explorer.html>

localhost:9870/explorer.html#/

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities

Browse Directory

/ Go!

Show 25 entries Search:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwx-----	hadoopkhai	supergroup	0 B	Feb 23 16:48	0	0 B	tmp
drwxr-xr-x	hadoopkhai	supergroup	0 B	Feb 23 16:42	0	0 B	user

Showing 1 to 2 of 2 entries Previous 1 Next

6. Xem các nhật ký làm việc của Hadoop

<http://localhost:9870/logs/> và chọn mở một file.

localhost:9870/logs/

Directory: /logs/

Name ↑	Last Modified	Size
hadoop-hadoopkhai-datanode-khai-master.log	Mar 2, 2025, 7:33:35 AM	342,631 bytes
hadoop-hadoopkhai-datanode-khai-master.out	Mar 2, 2025, 7:33:33 AM	818 bytes
hadoop-hadoopkhai-datanode-khai-master.out.1	Mar 2, 2025, 7:32:06 AM	818 bytes
hadoop-hadoopkhai-datanode-khai-master.out.2	Mar 2, 2025, 7:30:00 AM	818 bytes
hadoop-hadoopkhai-datanode-khai-master.out.3	Feb 23, 2025, 5:28:08 PM	818 bytes
hadoop-hadoopkhai-datanode-khai-master.out.4	Feb 23, 2025, 4:30:08 PM	818 bytes
hadoop-hadoopkhai-datanode-khai-master.out.5	Feb 23, 2025, 4:23:18 PM	818 bytes
hadoop-hadoopkhai-datanode-tqkhai.log	Feb 23, 2025, 11:31:46 AM	199,450 bytes
hadoop-hadoopkhai-datanode-tqkhai.out	Feb 23, 2025, 11:02:21 AM	818 bytes
hadoop-hadoopkhai-datanode-tqkhai.out.1	Feb 23, 2025, 10:52:27 AM	818 bytes
hadoop-hadoopkhai-datanode-tqkhai.out.2	Feb 23, 2025, 10:51:31 AM	818 bytes
hadoop-hadoopkhai-datanode-tqkhai.out.3	Feb 23, 2025, 10:50:30 AM	818 bytes

```
localhost:9870/logs/hadoo x +
localhost:9870/logs/hadoop-hadoopkhai-datanode-khai-master.log
2025-02-23 15:27:44,408 INFO org.apache.hadoop.hdfs.server.DataNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting DataNode
STARTUP_MSG: host = khai-master/192.168.111.1
STARTUP_MSG: args = []
STARTUP_MSG: version = 3.3.4
STARTUP_MSG: classpath = /home/hadoopkhai/hadoop/etc/hadoop:/home/hadoopkhai/hadoop/share/hadoop/common/lib/paranamer-2.3.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/hadoop-shaded-guava-1.1.1.jar:/
home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-util-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/listenablefuture-9999.0-empty-to-avoid-conflict-with-guava.jar:/home/hadoopkhai/
hadoop/share/hadoop/common/lib/commons-collections-3.2.2.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-common-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-core-1.0.1.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/kerb-server-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/token-provider-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jettison-1.1.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/commons-codec-1.15.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-crypto-1.0.1.jar:/
home/hadoopkhai/hadoop/share/hadoop/common/lib/libusb-jose-jwt-9.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-beanutils-1.9.4.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-
identity-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/cip-annotations-1.0.1.jar:/home/hadoopkhi/hadoop/share/hadoop/common/lib/stax2-api-4.2.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/
re2j-1.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jul-to-slf4j-1.7.36.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-
util-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/hadoop-shaded-protobuf-3-7-1.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-configuration2-2.11.1.jar:/home/hadoopkhai/hadoop/
share/hadoop/common/lib/kerby-pkix-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/httpcore-4.4.13.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/zookeeper-jute-3.5.6.jar:/home/hadoopkhai/hadoop/
share/hadoop/common/lib/guava-27.0-jre.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-logging-1.1.3.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/audience-annotations-0.5.0.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/kerb-client-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-core-asl-1.9.13.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/accessors-
smart-2.4.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/reload4j-1.2.22.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-
text-1.4.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/failureaccess-1.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-server-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/
lib/commons-cli-1.2.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jakarta.activation-api-1.2.1.jar:/home/hadoopkhai/hadoop/share/
hadoop/common/lib/slf4j-api-1.7.36.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jersey-servlet-1.19.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar:/home/hadoopkhai/hadoop/
share/hadoop/common/lib/commons-lang3-3.12.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/snappy-java-1.1.8.2.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/hadoop-auth-3.3.4.jar:/home/hadoopkhai/
hadoop/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/netty-3.10.6.Final.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-core-2.12.7.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/commons-net-3.6.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/javax-servlet-api-3.1.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/httpclient-4.5.13.jar:/
home/hadoopkhai/hadoop/share/hadoop/common/lib/avro-1.7.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/woodstox-core-5.3.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jsr305-3.0.2.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/jersey-core-1.19.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-math3-3.1.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-simplekdc-1.0.1.jar:/
home/hadoopkhai/hadoop/share/hadoop/common/lib/curator-recipes-4.2.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerby-util-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-
annotations-2.12.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-databind-2.12.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jersey-json-1.19.jar:/home/hadoopkhai/hadoop/share/hadoop/
common/lib/animal-sniffer-annotations-1.17.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/json-smart-2.4.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/gson-2.8.9.jar:/home/hadoopkhai/hadoop/share/
hadoop/common/lib/kerby-xdr-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-io-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/curator-framework-4.2.0.jar:/home/hadoopkhai/
hadoop/share/hadoop/common/lib/kerby-asn1-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/curator-client-4.2.0.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-util-
ajax-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-jaxrs-1.9.13.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-daemon-1.0.13.jar:/home/hadoopkhai/hadoop/share/hadoop/
common/lib/jetty-webapp-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-security-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerby-config-1.0.1.jar:/home/
hadoopkhai/hadoop/share/hadoop/common/lib/jetty-http-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-xml-1.9.13.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/asn-5.0.4.jar:/
home/hadoopkhai/hadoop/share/hadoop/common/lib/metrics-core-3.2.4.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/j2objc-annotations-1.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/kerb-
admin-1.0.1.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/commons-compress-1.21.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/dnsjava-2.1.7.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/
hadoop-annotations-3.3.4.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jetty-xsl-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jackson-mapper-asl-1.9.13.jar:/home/hadoopkhai/hadoop/
share/hadoop/common/lib/zookeeper-3.5.6.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/checker-qual-2.5.2.jar:/home/hadoopkhai/hadoop/share/hadoop/common/lib/jersey-server-1.19.jar:/home/hadoopkhai/hadoop/
hadoopkhai/hadoop/share/hadoop/common/hadoop-common-3.3.4-tests.jar:/home/hadoopkhai/hadoop/share/hadoop/common/hadoop-registry-3.3.4.jar:/home/hadoopkhai/hadoop/share/hadoop/kms-3.3.4.jar:/home/
hadoopkhai/hadoop/share/hadoop/hadoop-hdfs-3.3.4.jar:/home/hadoopkhai/hadoop/share/hadoop/hdfs:/home/hadoopkhai/hadoop/share/hadoop/hdfs:/home/hadoopkhai/hadoop/share/hadoop/
hdfs/lib/paranamer-2.3.jar:/home/hadoopkhai/hadoop/share/hadoop/hdfs/lib/hadoop-shaded-guava-1.1.1.jar:/home/hadoopkhai/hadoop/share/hadoop/hdfs/lib/jetty-util-9.4.43.v20210629.jar:/home/hadoopkhai/hadoop/
share/hadoop/hdfs/lib/listenablefuture-9999.0-empty-to-avoid-conflict-with-guava.jar:/home/hadoopkhai/hadoop/share/hadoop/hdfs/lib/netty-codec-smtp-4.1.77.Final.jar:/home/hadoopkhai/hadoop/share/hadoop/hdfs/
```

7. Xem đường dẫn thư mục cài đặt Hadoop

\$ cd \$HADOOP_HOME

8. Xem file cấu hình hdfs và map-reduce

\$ cd etc/hadoop

\$ sudo nano core-site.xml

\$ sudo nano hdfs-site.xml

\$ sudo yarn-site.xml

\$ sudo yarn mapred-site.xml

\$ cd

9. (hdfs) Xem không gian dung lượng đĩa

Usage: hadoop fs -df [-h] URI [URI ...] (displays free space)

\$ hdfs dfs -df -h

10.(hdfs) Liệt kê danh sách tập tin, thư mục

\$ hdfs dfs -ls /

```
hadoopkhai@khai-master:~$ hdfs dfs -ls /
Found 2 items
drwx----- - hadoopkhai supergroup          0 2025-02-23 16:48 /tmp
drwxr-xr-x - hadoopkhai supergroup          0 2025-02-23 16:42 /user
hadoopkhai@khai-master:~$
```

11.(hdfs) Tìm kiếm thông tin thư mục, tập tin

Usage: hadoop fs -find <path> ... <expression> ...

Finds all files that match the specified expression and apply selected actions to them. If no path is specified then defaults to the current working directory. If no expression is specified then defaults to -print.

```
$ hdfs dfs -find / -name tmp
```

12.(hdfs) Tạo thư mục mới

Usage: `hadoop fs -mkdir [-p] <paths>`

Takes path uri's as argument and creates directories.

```
$ hdfs dfs -mkdir /khai
```

Sau khi tạo thư mục mới thành công, dùng lệnh `$ hdfs dfs -ls /`

hoặc dùng

<http://localhost:9870/explorer.html> để xem kết quả tạo thư mục mới.

13.(hdfs) Tạo mới tập tin kích thước 0 bytes trong thư mục cho trước.

Usage: `hdfs dfs -touch [-a] [-m] [-t TIMESTAMP] [-c] URI [URI ...]`

Updates the access and modification times of the file specified by the URI to the current time. If the file does not exist, then a zero length file is created at URI with current time as the timestamp of that URI.

Usage: `hdfs dfs -touchz URI [URI ...]`

Create a file of zero length. An error is returned if the file exists with non-zero length.

```
$ hdfs dfs -touchz /khai/sample.txt
```

14.(hdfs) Kiểm tra kích thước tập tin

Usage: `hdfs dfs -du -s <args>` (displays a summary of file lengths.)

```
$ hdfs dfs -du -s /khai/sample.txt
```

```
hadoopkhai@khai-master:~$ hdfs dfs -du -s /khai/sample.txt
0    0    /khai/sample.txt
hadoopkhai@khai-master:~$
```

15.(hdfs) Nối các tập tin với nhau

Usage: `hdfs dfs -concat <target file> <source files>`

Concatenate existing source files into the target file. Target file and source files should be in the same directory.

```
$ hdfs dfs -concat /khai/sample.txt /khai/sample1.txt
```

Lưu ý: cần khởi tạo tập tin /khai/sample1.txt trước

16.(hdfs) Xem thời gian chỉnh sửa cuối cùng của thư mục

Usage: `hadoop fs -stat [format] <path> ...`

Print statistics about the file/directory at <path> in the specified format. Format accepts permissions in octal (%a) and symbolic (%A), filesize in bytes (%b), type (%F), group name of owner (%g), name (%n), block size (%o), replication (%r), user name of owner(%u), access date(%x, %X), and modification date (%y, %Y). %x and %y show UTC date as “yyyy-MM-dd HH:mm:ss”, and %X and %Y show milliseconds since January 1, 1970 UTC. If the format is not specified, %y is used by default.

```
$ hdfs dfs -stat /
```

17.(local–linux) Tạo mới thư mục

```
$ mkdir test
```

18.(local–linux) Liệt kê thư mục và tập tin

```
$ ls -l
```

19.(local–linux) Tạo mới tập tin

```
$ cat > test/data.txt
```

Hello

Tran Quang Khai

ctrl + D

20.(local–linux) Xem lại nội dung tập tin

```
$ cat /test/data.txt
```

21.(hdfs) Đưa dữ liệu từ local lên hdfs (có thể dùng lệnh put để thay thế)

Usage: `hdfs dfs -put [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] [ - | <localsrc> ...] <dst>`

Copy single src, or multiple srcs from local file system to the destination file system. Also reads input from stdin and writes to destination file system if the source is set to “-”. Copying fails if the file already exists, unless the -f flag is given

```
$ hdfs dfs -copyFromLocal /home/hadoopkhai/test/data.txt /khai
```

hoặc

```
$ hdfs dfs -put /home/hadoopkhai/test/data.txt /khai
```

22.(hdfs) Di chuyển dữ liệu từ local lên hdfs

Usage: `hdfs dfs -moveFromLocal <localsrc> <dst>` (similar to put command, except that the source localsrc is deleted after it’s copied)

```
$ hdfs dfs -moveFromLocal /home/hadoopkhai/test/data1.txt /khai
```

Lưu ý: cần tạo tập tin data1.txt trước

23.(hdfs) Xem kết quả copy dữ liệu lên hdfs

Usage: hdfs dfs -ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] <args>

\$ hdfs dfs -ls /khai

```
hadoopkhai@khai-master:~/test$ hdfs dfs -ls /khai
Found 2 items
-rw-r--r--  2 hadoopkhai supergroup      21 2025-03-02 07:56 /khai/data.txt
-rw-r--r--  2 hadoopkhai supergroup       0 2025-03-02 07:49 /khai/sample.txt
hadoopkhai@khai-master:~/test$
```

24.(hdfs) Đọc nội dung và in nội dung của file ra màn hình

\$ hdfs dfs -cat /khai/data.txt

25.(hdfs) Lấy nội dung của tập tin và xuất ra dạng text

Usage: hdfs dfs -text <src>

Takes a source file and outputs the file in text format. The allowed formats are zip and TextRecordInputStream.

\$ hdfs dfs -text /khai/data.txt

26.(hdfs) Đếm số lượng thư mục, tập tin và dung lượng của đường dẫn

Usage: hadoop fs -count [-q] [-h] [-v] [-x] [-t [<storage type>]] [-u] [-e] [-s] <paths>

Count the number of directories, files and bytes under the paths that match the specified file pattern. Get the quota and the usage. The output columns with -count are: DIR_COUNT, FILE_COUNT, CONTENT_SIZE, PATHNAME

\$ hdfs dfs -count /khai

```
hadoopkhai@khai-master:~$ hdfs dfs -count /khai
      1      2      21 /khai
hadoopkhai@khai-master:~$
```

27.(hdfs) Thêm dữ liệu tập tin từ local lên hdfs

Usage: hdfs dfs -appendToFile <localsrc> ... <dst>

Append single src, or multiple srcs from local file system to the destination file system.

Also reads input from stdin and appends to destination file system.

\$ hdfs dfs -appendToFile /home/hadoopkhai/test/data1.txt /khai/data.txt

Lưu ý: cần tạo tập tin data1.txt ở local và data.txt ở hdfs trước

28.(hdfs) Xóa tập tin trên hdfs

\$ hdfs dfs -rm /khai/sample.txt


```
hadoopkhai@khai-master:~$ hdfs dfs -rm /khai/sample.txt
Deleted /khai/sample.txt
hadoopkhai@khai-master:~$
```

29.(hdfs) Sao chép tập tin đến đường dẫn lưu trữ trong cùng hdfs.

\$ hdfs dfs -cp /tmp /khai

```
hadoopkhai@khai-master:~$ hdfs dfs -ls /khai
Found 2 items
-rw-r--r--  2 hadoopkhai supergroup      21 2025-03-02 07:56 /khai/data.txt
drwxr-xr-x  - hadoopkhai supergroup      0 2025-03-02 08:00 /khai/tmp
hadoopkhai@khai-master:~$
```

30.(hdfs) Di chuyển tập tin đến đường dẫn lưu trữ hdfs.

Usage: hdfs dfs -mv URI [URI ...] <dest> (moves files from source to destination. This command allows multiple sources as well in which case the destination needs to be a directory. Moving files across file systems is not permitted)

\$ hdfs dfs -mv /khai/data.txt /newkhai

Lưu ý: tạo thư mục newkhai trước

```
hadoopkhai@khai-master:~$ hdfs dfs -ls /newkhai
Found 1 items
-rw-r--r--  2 hadoopkhai supergroup      21 2025-03-02 07:56 /newkhai/data.txt
hadoopkhai@khai-master:~$ █
```

31.(hdfs) Xóa thư mục newkhai

\$ hdfs dfs -rmr -r /newkhai (dùng để xóa thư mục không rỗng)

32.(hdfs) Đưa dữ liệu từ hdfs về local (có thể dùng lệnh get để thay thế)

Usage: hdfs dfs -get [-ignoreCrc] [-crc] [-p] [-f] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst> (copy files to the local file system. Files that fail the CRC check may be copied with the -ignoreCrc option. Files and CRCs may be copied using the -crc option.)

\$ hdfs dfs -copyToLocal /khai/tmp /home/hadoopkhai/test

hoặc

\$ hdfs dfs -get /khai/tmp /home/hadoopkhai/test

33.(Local) Kiểm tra kết quả copy tmp về local:

\$ ls -l test

```
hadoopkhai@khai-master:~$ ls test
data.txt  tmp
hadoopkhai@khai-master:~$ █
```

34.(hdfs) Xóa hết tập tin thư mục trong đường dẫn

Usage: `hdfs dfs -rm [-f] [-r | -R] [-skipTrash] [-safely] URI [URI ...]` (delete files specified as args)

`$ hdfs dfs -rm -r /khai`

35. (hdfs) Xóa thư mục rỗng

Usage: `hdfs dfs -rmdir [--ignore-fail-on-non-empty] URI [URI ...]` (delete a directory)

Usage: `hdfs dfs -rm -r [-skipTrash] URI [URI ...]` (recursive version of delete)

`$ hdfs dfs -rmdir /khai`

hoặc

`$ hdfs dfs -rm -r /khai`

36. (hdfs) Làm sạch thùng rác

Usage: `hdfs dfs -expunge [-immediate] [-fs <path>]` (permanently delete files in checkpoints older than the retention threshold from trash directory and create new checkpoint.)

`$ hdfs dfs -expunge`

37. (hdfs) Xem cấu trúc câu lệnh

`$ hdfs dfs -usage mkdir`

38. (hdfs) Hiện thị chương trình trợ giúp

`$ hdfs dfs -help`

Bên cạnh đó còn một số lệnh sau:

chgrp

Usage: `hadoop fs -chgrp [-R] GROUP URI [URI ...]`

Change group association of files. The user must be the owner of files, or else a super-user. Additional information is in the [Permissions Guide](#).

chmod

Usage: `hadoop fs -chmod [-R] <MODE[,MODE]... | OCTALMODE> URI [URI ...]`

Change the permissions of files. With -R, make the change recursively through the directory structure. The user must be the owner of the file, or else a super-user. Additional information is in the [Permissions Guide](#).

chown

Usage: `hadoop fs -chown [-R] [OWNER][:[GROUP]] URI [URI ]`

Change the owner of files. The user must be a super-user. Additional information is in the [Permissions Guide](#).

head

Usage: `hadoop fs -head URI`

Displays first kilobyte of the file to stdout.

tail

Usage: `hadoop fs -tail [-f] URI`

Displays last kilobyte of the file to stdout.

truncate

Usage: `hadoop fs -truncate [-w] <length> <paths>`

Truncate all files that match the specified file pattern to the specified length.

Options:

Nguồn tham khảo:

<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-common/FileSystemShell.html>